

Chapter 11

Force and Pressure

Delhi Public School · Bokaro Steel City · Subject: Science

11.1 What is Force?

In everyday life we push or pull objects to make them move, stop, or change direction. A **force** is a push or pull exerted on an object. It can change the state of rest or motion of a body, alter its speed, or change its shape.

DEFINITION

Force is any interaction that, when unopposed, changes the motion of an object. Force is a *vector quantity* — it has both magnitude and direction. The SI unit of force is the **Newton (N)**.

Effects of Force

A force applied on an object may produce one or more of the following effects:

- Make a stationary object move (e.g., kicking a football at rest).
- Stop a moving object (e.g., applying brakes on a bicycle).
- Change the speed of a moving object.
- Change the direction of motion of an object.
- Change the shape or size of an object (e.g., squeezing a rubber ball).

EVERYDAY EXAMPLE

When you push a door open, you apply a force. The door, initially at rest, begins to move — a clear demonstration that force can cause motion.

11.2 Types of Forces

11.2.1 Contact Forces

Forces that act only when two objects are in physical contact are called **contact forces**.

- **Muscular Force:** Force exerted by muscles. Examples — lifting a bag, running.
- **Friction:** A contact force that opposes relative motion between surfaces. Friction allows us to walk without slipping.
- **Normal Force:** The perpendicular force a surface exerts on an object resting on it.
- **Spring Force:** Force exerted by a compressed or stretched spring.

11.2.2 Non-Contact Forces

Forces that act between objects without physical contact are called **non-contact forces**.

- **Gravitational Force:** Attractive force between any two masses. Earth's gravity pulls every object toward its centre.
- **Magnetic Force:** Force between magnets or between a magnet and a magnetic material. Like poles repel; unlike poles attract.
- **Electrostatic Force:** Force between electrically charged objects. Like charges repel; unlike charges attract.

Force Type	Contact / Non-contact	Example
Muscular	Contact	Lifting a book
Friction	Contact	Braking a bicycle
Gravitational	Non-contact	Apple falling from tree
Magnetic	Non-contact	Compass needle aligning
Electrostatic	Non-contact	Comb attracting paper bits

11.3 Resultant Force

When more than one force acts on an object at the same time, the combined effect is described by the **resultant (net) force**.

- If two forces act in the **same direction**, they add up: $\text{Net Force} = F_1 + F_2$.
- If two forces act in **opposite directions**, they partially cancel: $\text{Net Force} = F_1 - F_2$ (in the direction of the larger force).
- If the net force is **zero**, the object remains in its current state (rest or uniform motion) — a condition called *equilibrium*.

NUMERICAL EXAMPLE

Two children push a box from the same side — one with 30 N, the other with 20 N. Net force = $30 + 20 = 50 \text{ N}$ in that direction. If they push from opposite sides, net force = $30 - 20 = 10 \text{ N}$ toward the child pushing with 30 N.

11.4 Pressure

The same force can have very different effects depending on the area over which it acts. **Pressure** is defined as the force acting perpendicularly per unit area of a surface.

$$\text{Pressure (P)} = \text{Force (F)} \div \text{Area (A)} \quad | \quad \text{Unit : Pascal (Pa)} = \text{N/m}^2$$

A sharp knife cuts easily because the same force is concentrated on a very small area, producing high pressure. A wide ski distributes your weight over a large area, reducing pressure on snow so you don't sink.

11.4.1 Pressure in Liquids and Gases

Liquids and gases exert pressure in **all directions**. The pressure exerted by a liquid column increases with depth:

$$\text{Liquid Pressure} = \rho \times g \times h \quad | \quad \rho = \text{density}, g = 9.8 \text{ m/s}^2, h = \text{depth}$$

This is why deep-sea fish are adapted to withstand enormous pressures, and why dams are built thicker at the base.

Did You Know? Atmospheric pressure at sea level is approximately 101,325 Pa (about 101 kPa). This pressure decreases as you go higher — that's why it is harder to breathe at

high altitudes!

11.4.2 Atmospheric Pressure

The air around us has weight. The pressure exerted by the atmosphere on everything it touches is called **atmospheric pressure**. It is measured using a **barometer**. At sea level, the atmosphere supports a column of mercury 76 cm tall.

11.5 Applications of Pressure

- **Cutting tools** (knife, scissors, chisel) have sharp edges to increase pressure.
- **Wide tyres** on tractors spread weight over soft soil, reducing pressure and preventing sinking.
- **Nails and needles** have pointed tips to penetrate surfaces easily.
- **Hydraulic machines** (car brakes, jacks) use liquid pressure to multiply force over a larger piston area.
- **Syringes** use pressure difference to draw and inject fluids.

Chapter Summary

1. A **force** is a push or pull that changes the state of motion or shape of an object.
2. Forces are classified as **contact forces** (muscular, friction, normal) and **non-contact forces** (gravitational, magnetic, electrostatic).
3. The **net force** is the vector sum of all forces on an object; zero net force means equilibrium.
4. **Pressure** = Force ÷ Area; SI unit is the Pascal (Pa).
5. Liquids and gases exert pressure in all directions; liquid pressure increases with depth.

6. **Atmospheric pressure** is the weight of the air column above a surface, measurable with a barometer.